

AIAA 2290: Ethics, Privacy and Security in AI

Security in Computer Vision and Robotics

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The Hong Kong University of Science and Technology (Guangzhou)

2025 Fall



1 Security Risks in Computer Vision and Robotics

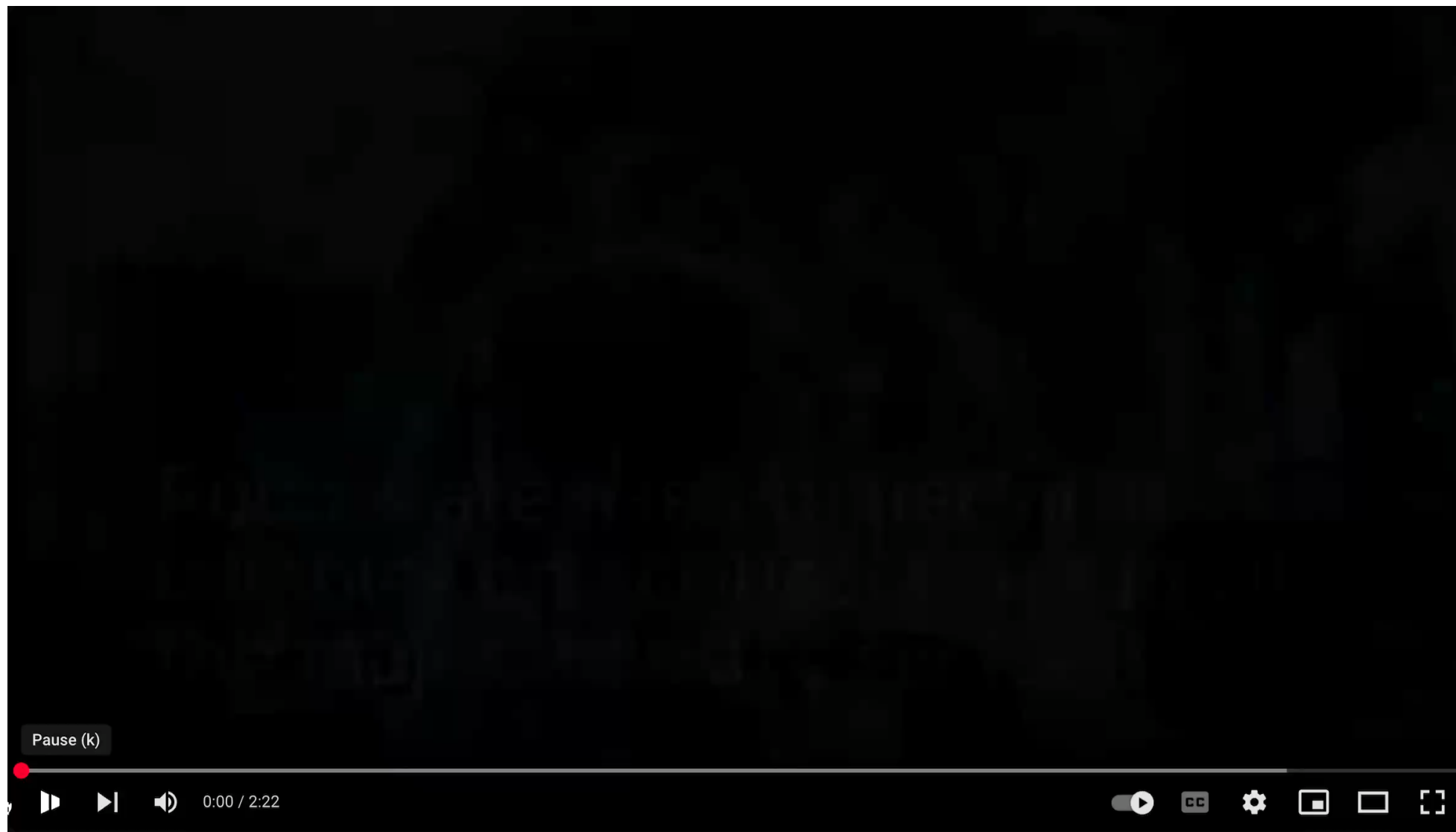
2 Defense in Computer Vision and Robotics

3 Security of Sensor Networks

4 Practice:

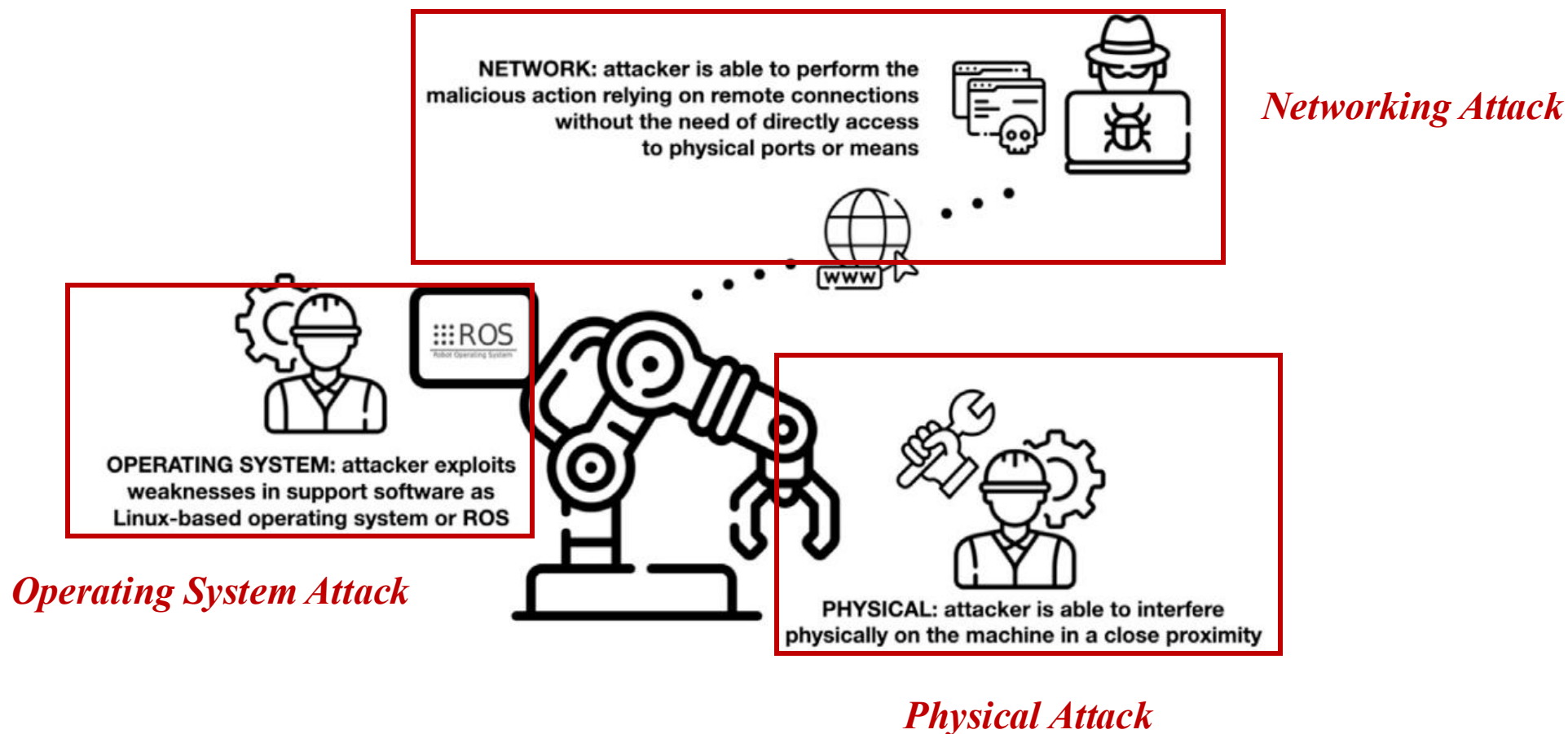


Cyber-attacks on CV and robotics systems





Types of attacks on robotic platforms





Types of attacks on robotic platforms

Physical Attack

*Physical layer attack means the **hardware-based attacks***

*The effects of hardware attacks should be particularly considered, by taking into account the **specific sensors** and actuators applied in the robotics field*



Types of attacks on robotic platforms

Physical Attack

An important question regarding this matter is the communication between the control system and the peripherals. In this context, the **Inter-Integrated Circuit (I2C)** protocol is a commonly used mechanism to establish this communication process. In fact, the use of I2C low level controllers in mobile robotics has spread in the last years. I2C is a synchronous protocol, where the **clock signal** plays a relevant role. Clock synchronization heavily depends on the applied synchronization protocol and algorithm



Types of attacks on robotic platforms

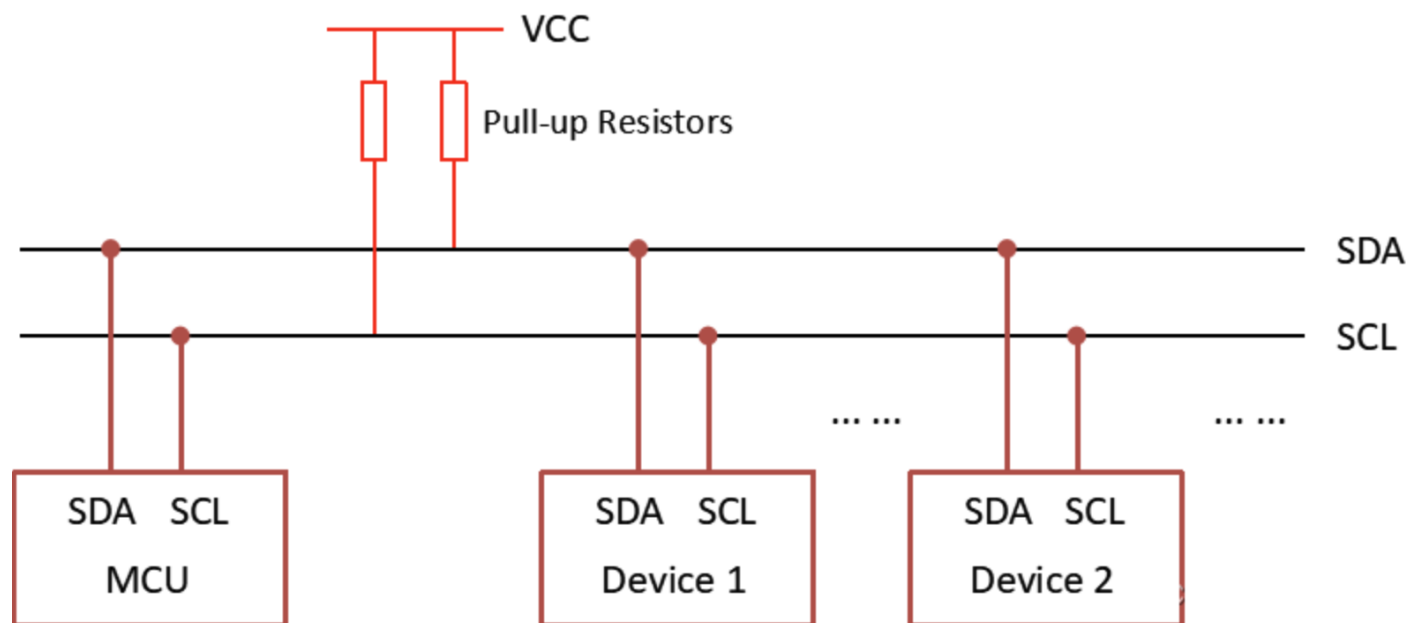
Physical Attack

Inter-Integrated Circuit (I2C)

I2C has **two buses**, a bi-directional serial data line, SDA, and a serial clock line, SCL:

SDA (Serial data): Serial data line, used to transmit data;

SCL (Serial clock line): clock line, used to control the **timing of data transmission**.





Types of attacks on robotic platforms

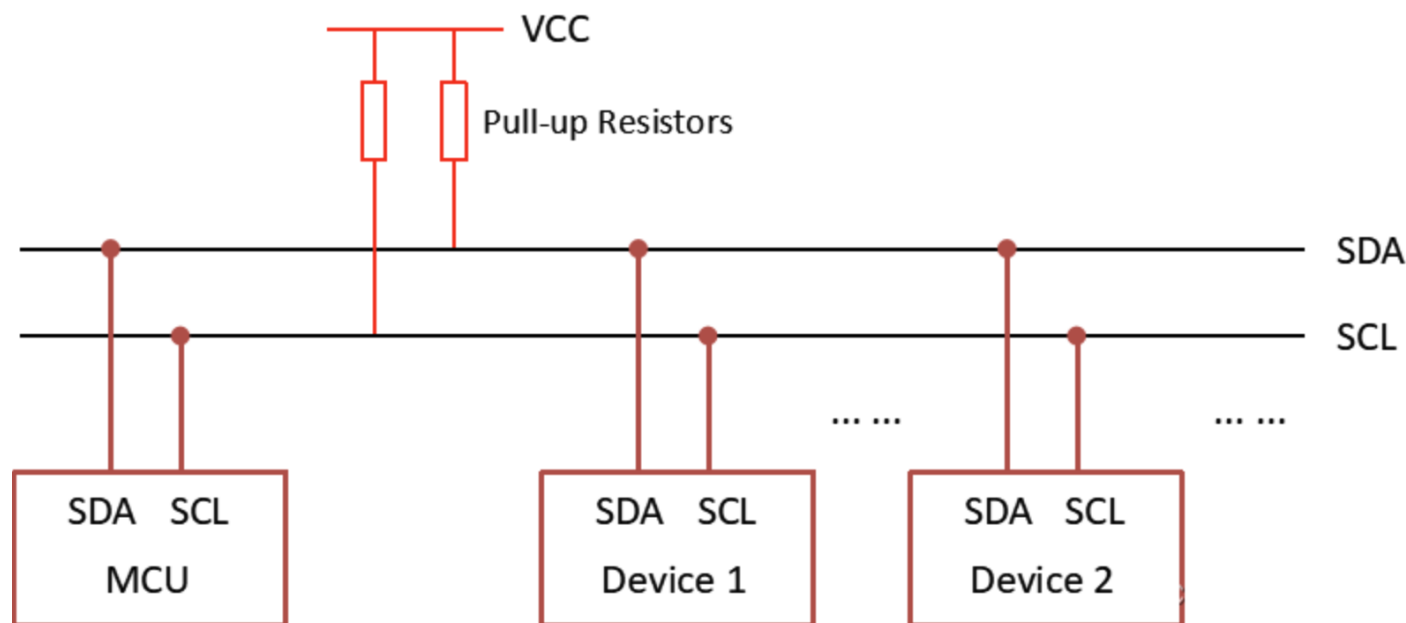
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Attack to the clock signal



Types of attacks on robotic platforms

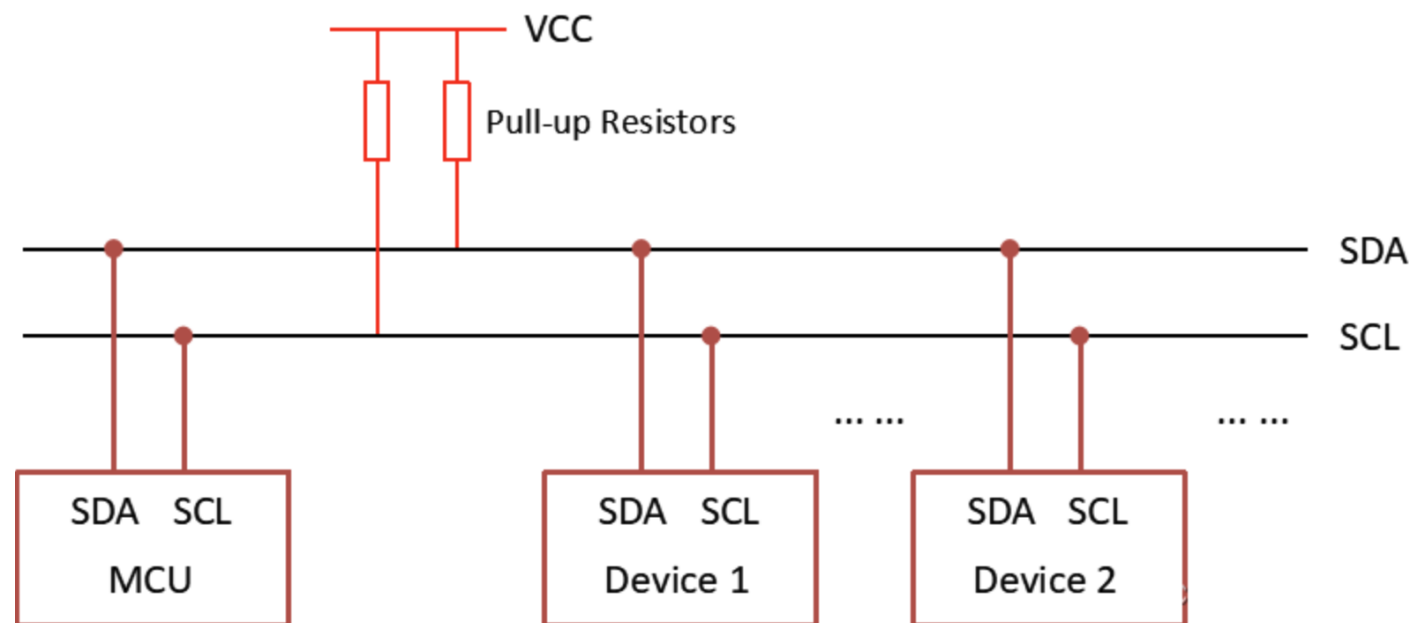
Physical Attack

Inter-Integrated Circuit (I2C)

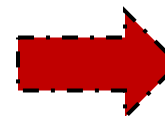
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Attack to the clock signal



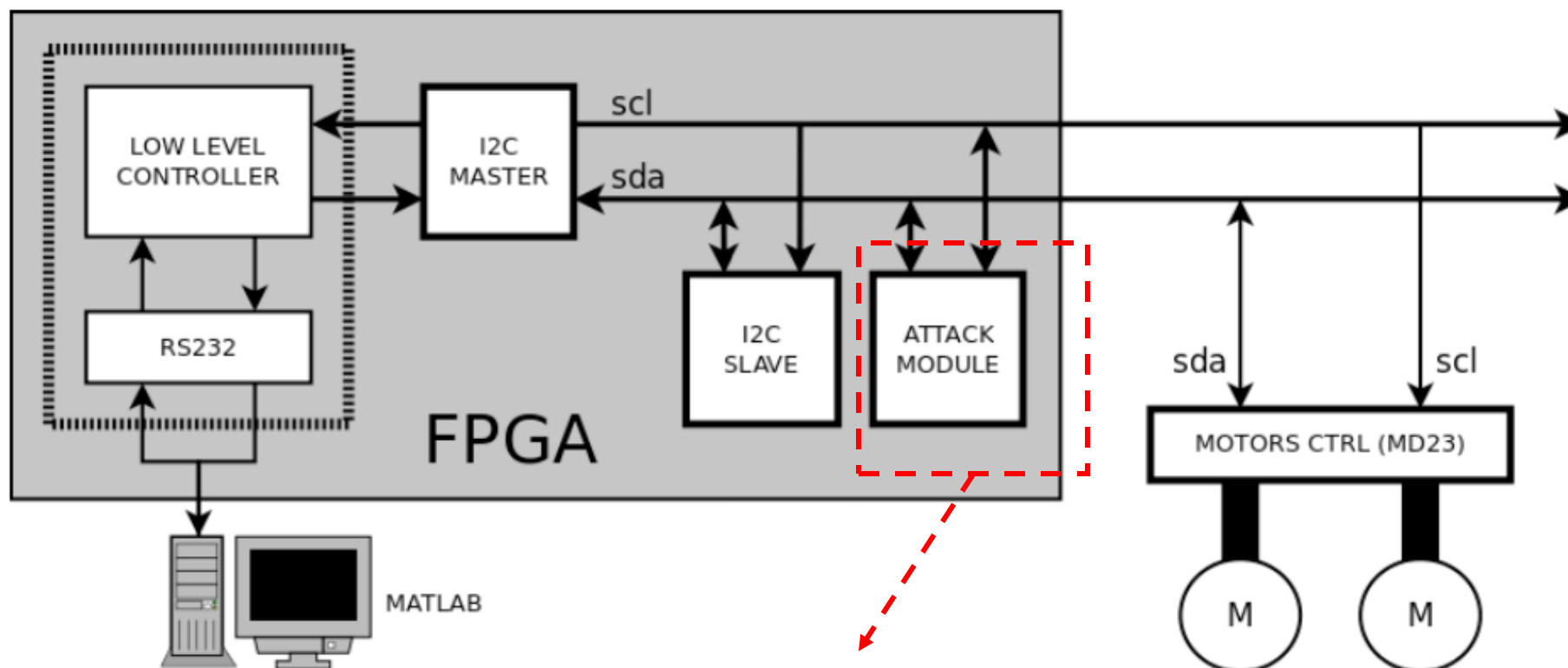
Timing error



Types of attacks on robotic platforms

Physical Attack

Attacking Framework



Add attacking clock signal



Types of attacks on robotic platforms

Physical Attack

**Nominal
Clock Signal:**



+

**Attack
Clock Signal:**



+

**Monitoring
Clock Signal:**

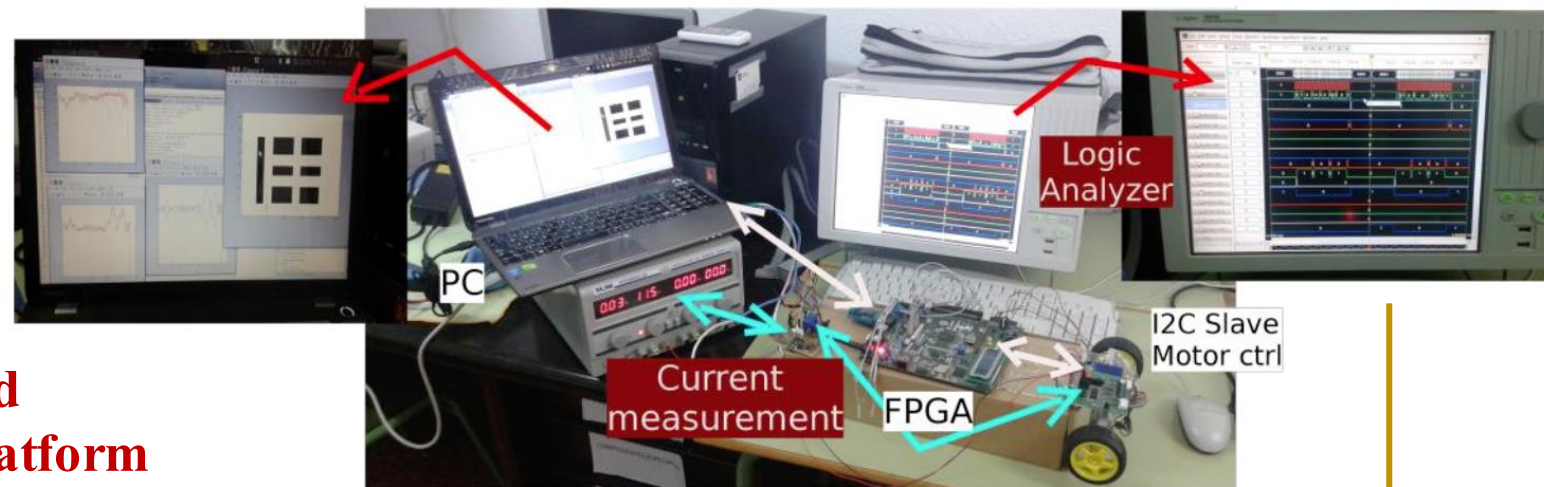




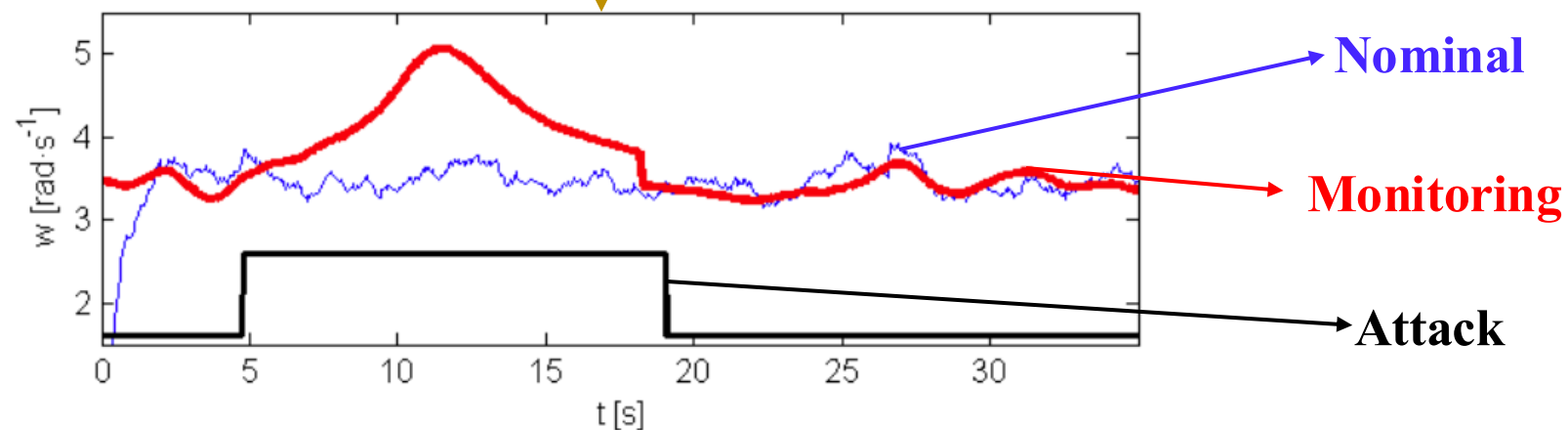
Types of attacks on robotic platforms

Physical Attack

**Real-world
experimental platform**



Output Signals:





Types of attacks on robotic platforms

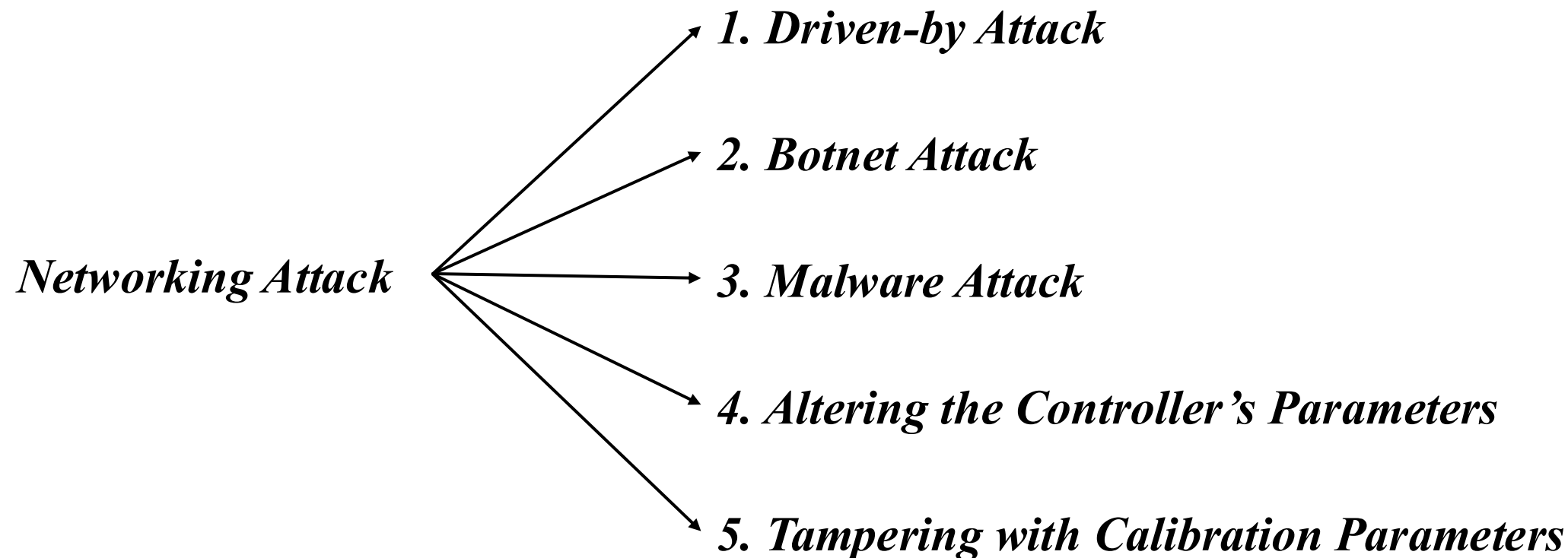
Networking Attack

*Performing malicious actions by **remote connections** without the need to directly access to physical ports is called a “**Network Attack**”, which expands the attacker’s possibilities of actions effectively*



Types of attacks on robotic platforms

Networking Attack





Types of attacks on robotic platforms

Networking Attack: Driven-by attack

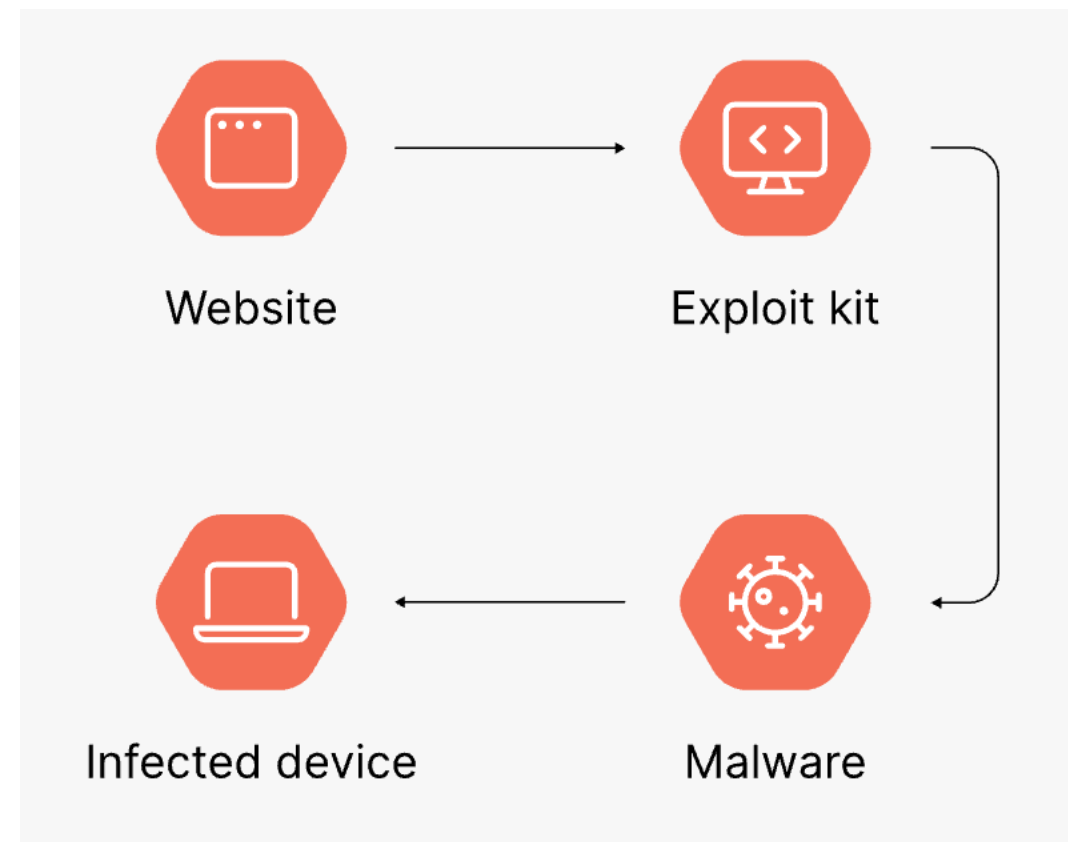
A drive-by attack, also known as a drive-by download attack, refers to a *cyberattack in which a malicious script causes a program to download and install itself on a user device*, without explicit permission from the user. It can happen on any user device, running any operating system. Often, these attacks occur when the user navigates to and browses a compromised web page.



Types of attacks on robotic platforms

Networking Attack: Driven-by attack

Drive-by attacks often use **exploit kits** to launch the automatic download. Exploit kits are malicious pieces of software, created by hackers to identify **vulnerabilities** in a device, web browser, or web-based app. These weaknesses are then used to launch the automatic download process and carry out the attack. Drive-by attacks are designed to *infect devices, steal information, and/or cause damage to data.*





Types of attacks on robotic platforms

Networking Attack: Driven-by attack

Drive-by attacks are named as such because the download occurs silently, as they pass by, leaving users mystified as to how their devices was infected. There are **two main methods** through which drive-by downloads can infect a user device:

- *Without authorization*
- *With authorization under false pretences*



Types of attacks on robotic platforms

Networking Attack: Driven-by attack

Without authorization

In these cases, there is **no direct user action** which causes the download to begin. In other words, *the attack is launched when the user visits a compromised webpage, without the need for any interaction with the page, not even a single click.*

To create such an attack, the hacker injects malicious code into the web page **by exploiting flaws in the website's security**. When the user first visits the page, the code identifies any security vulnerabilities in either the user's web browser or the user's device, and triggers the malware download through these vulnerabilities.



Types of attacks on robotic platforms

Networking Attack: Driven-by attack

With authorization under false pretences

Sometimes, there is a user action involved in a drive by download, but under false pretences. There are a few different ways this can be done, for example:

- A pop up advert could have **an 'X' in the corner** that disguises itself as a close button, but actually acts as a catalyst for **starting a malicious download once pressed**.
- A **link could appear legitimate**, but clicking on it could cause the download to begin.

All three examples involved a simple click or button press. These actions allow the attacker to claim that the user authorized the download, when in reality the user did not realize the implications of their actions, as the true intentions of the attacker were hidden. In this way, the hacker can get away with the download without being detected.



Types of attacks on robotic platforms

Networking Attack: Botnet attack

The Mirai botnet was first seen in August 2016 and has since been used to launch large DDoS attacks on websites, networks and other digital infrastructure. Mirai was published as **a source code by “Anna-senpai”** to a public and easily accessible forum. The malicious code allows an attacker to gain control of ***vulnerable IoT devices such as webcams, DVRs, IP cameras, and routers***. In early 2017, Krebs publicly named Josiah White and Paras Jha as the likely creators of Mirai botnet.

Security researchers estimate that ***there are millions of vulnerable IoT devices actively taking part in these coordinated attacks***. Mirai can also infect all devices connected to the same network, making it possible to create a large botnet capable of launching devastating attacks.



Types of attacks on robotic platforms

Networking Attack: Botnet attack

The Mirai botnet works by *scanning for vulnerable IoT devices* that have open ports or default usernames and passwords. Once it finds these vulnerable devices, it uses exploits to gain access and infects them with its malicious code. The infected device then joins the Mirai botnet which allows the attacker to send commands from a central server which is known as a *“command & control”* server (C&C). This C&C server can then be used to launch large-scale DDoS attacks on websites, networks and other digital infrastructure by using all of the bots in the Mirai Botnet at once.



Types of attacks on robotic platforms

Networking Attack: Malware attack

Malware attacks include injection of software which forces the system to misbehave. They can be injected by infiltrating the network or attacking nodes such as **edge devices or fog nodes**. The key aspect of malware is **its close resemblance to actual good software that makes detection hard**.

Malware attacks not only threaten information security, but since CPS involves physical actuation, they can cause real-world life-threatening damage. For example, **misbehaving surgical robots**, **bricking autonomous cars**, **causing disruption on a manufacturing floor with robots**, are few such attack vectors that have been studied. Hence, detection and subsequent isolation of malware are critical for robust and safe robotic systems



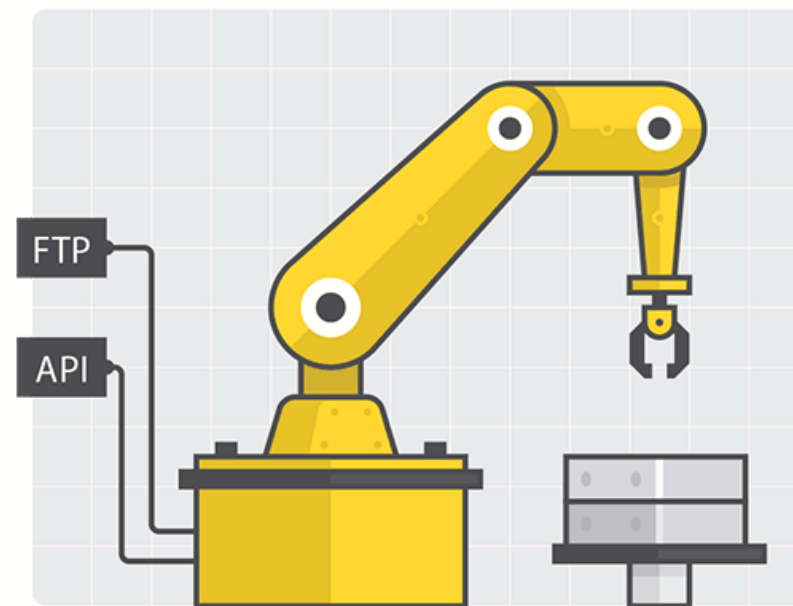
Types of attacks on robotic platforms

Networking Attack: Altering the Controller's Parameters

The attacker alters the control system so the robot moves unexpectedly or inaccurately, at the attacker's will.

Concrete Effects: Defective or modified products

Requirements Violated: Safety, Integrity, Accuracy





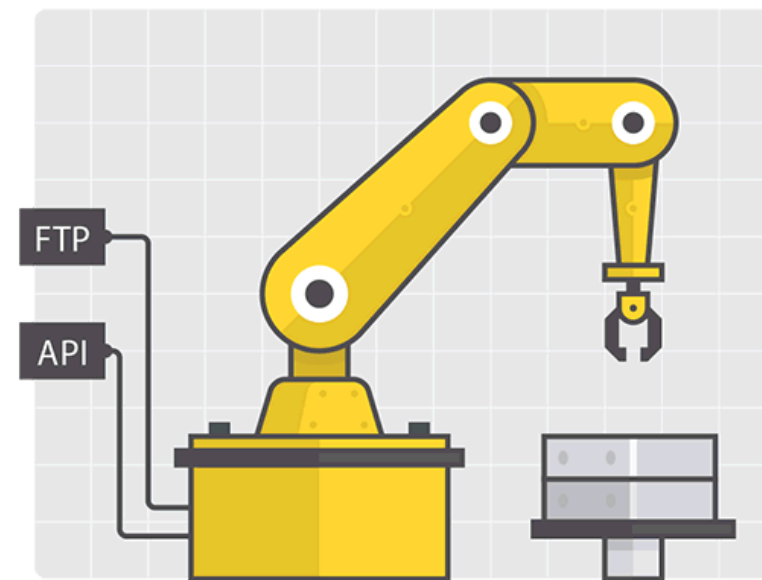
Types of attacks on robotic platforms

Networking Attack: Tampering with Calibration Parameters

The attacker changes the calibration to make the robot move or inaccurately, **unexpectedly** at the attacker's will.

Concrete Effects: Robot damages

Requirements Violated: Safety, Integrity, Accuracy





Types of attacks on robotic platforms

Operating System Attack

Operating system attacks exploit weaknesses in support software such as **Linux-based operating systems or ROS** which constitute the heart of several robots. Analyzing a total of 176 threats collected from the robot vulnerability database showed that **92.6%** are mainly software-related and proved that software in robotics systems poses a more critical threat in comparison with hardware components



Types of attacks on robotic platforms

Operating System Attack

Operating system attacks exploit weaknesses in support software such as **Linux-based operating systems or ROS** which constitute the heart of several robots. Analyzing a total of 176 threats collected from the robot vulnerability database showed that **92.6%** are mainly software-related and proved that software in robotics systems poses a more critical threat in comparison with hardware components

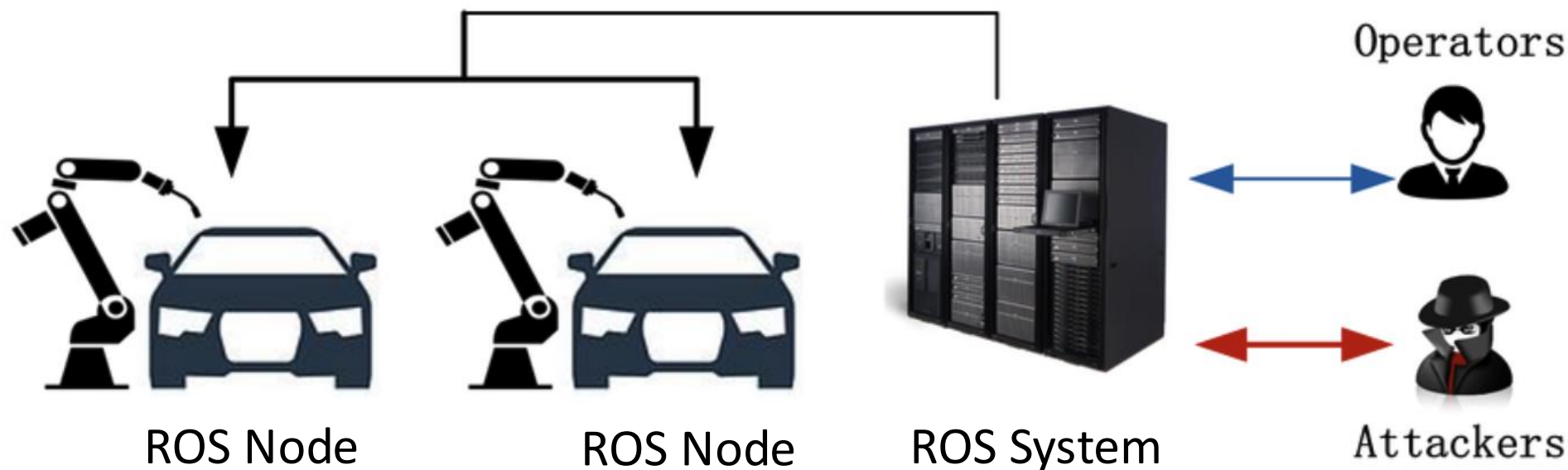




Types of attacks on robotic platforms

Operating System Attack

When dealing with **multi-robot active surveillance systems**, an **attacker** inside the network can easily send shutdown commands to manipulate or destroy any **ROS node** and publish adulterated messages manipulating velocity commands to misdirect a robot



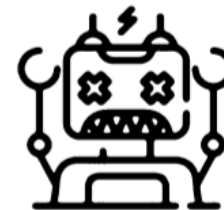


Attacks on Different Types of Robots and Their Effect



**impact motors and batteries
performance, crash the robot
into the wall, gains remote control**

Autonomous



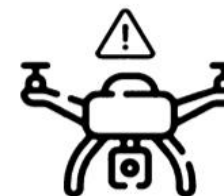
harm people around

Humanoid



**adjust home settings
switching off and on lights, playing
music, controlling remotely the robot**

Small Mobile



**crash the robot,
affect the robot response time
or shut it down**

Drones



Attacks on Different Types of Robots and Their Effect

Small Mobile



Sweeping Robot

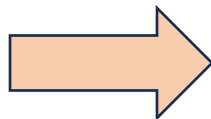


Delivery Robot



Attacks on Different Types of Robots and Their Effect

Small Mobile



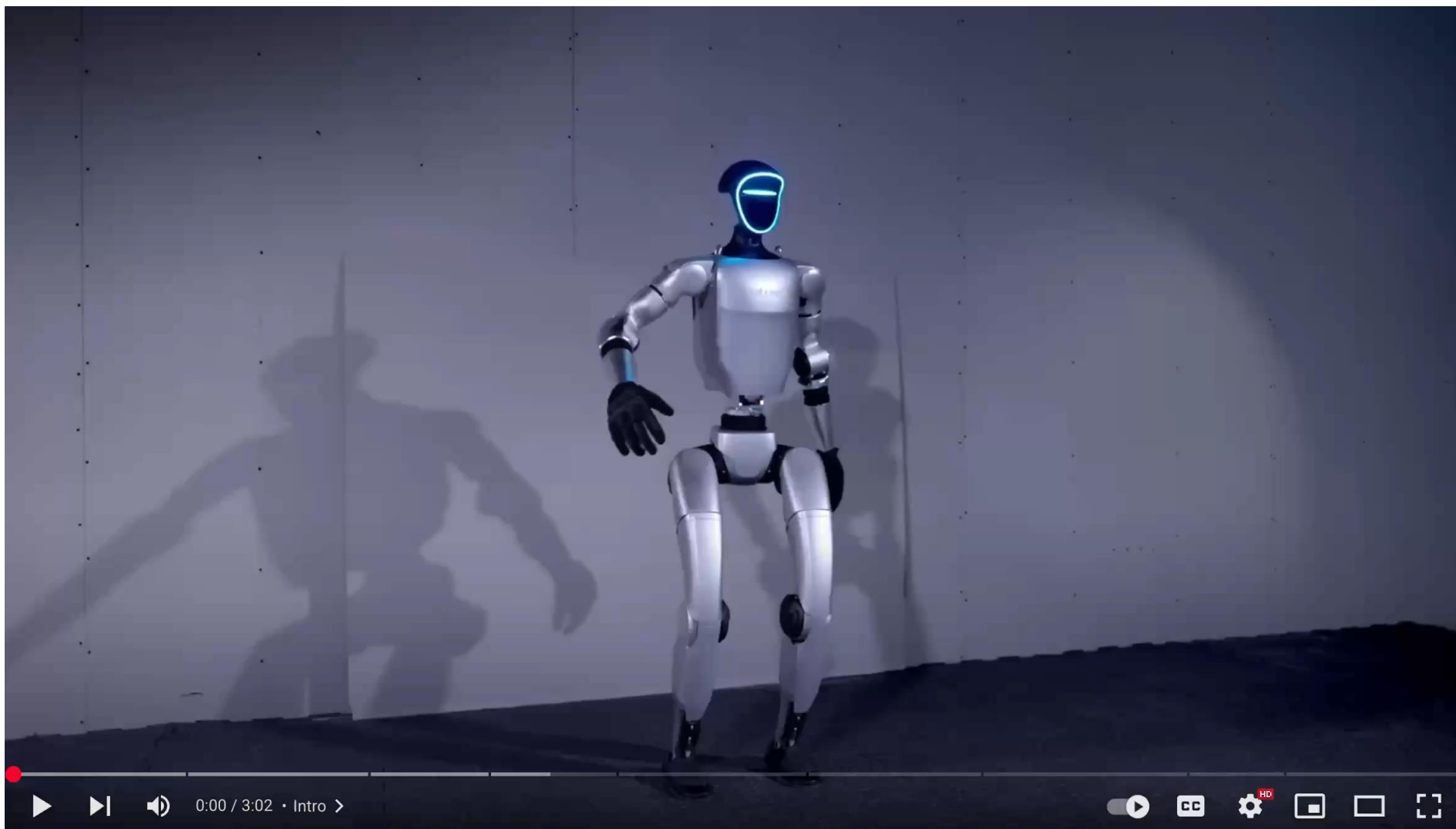
Potential Vulnerabilities

- **Network**
- **Encryption**
- **Software**
- **Hardware**



Attacks on Different Types of Robots and Their Effect

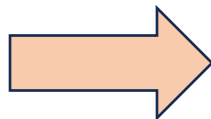
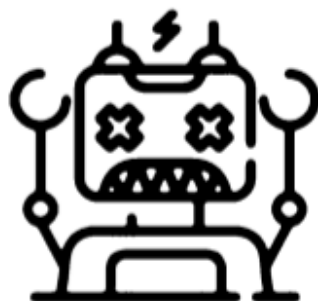
Humanoid





Attacks on Different Types of Robots and Their Effect

Humanoid



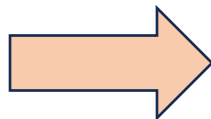
Potential Vulnerabilities

- **Network**
- **Encryption**
- **Authentication**
- **Authorization**



Attacks on Different Types of Robots and Their Effect

Autonomous



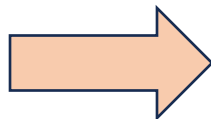
Potential Vulnerabilities

- **Network**
- **Data**
- **Controllers**



Attacks on Different Types of Robots and Their Effect

Drones



Potential Vulnerabilities

- **Network**
- **Encryption**
- **Authentication**
- **Authorization**
- **ROS**



Quiz

Q1: What does I2C stand for in electronics?

A) Integrated Interface Circuit

B) Internal Integrated Circuit

C) Inter-Integrated Circuit

D) Independent Integrated Circuit



Quiz

Q2: Which of the following is a key feature of the I2C communication protocol?

- A) It uses only one wire for communication.*
- B) It allows multiple devices to communicate using two wires.***
- C) It supports high-speed data transfer only.*
- D) It requires no clock signal for operation.*



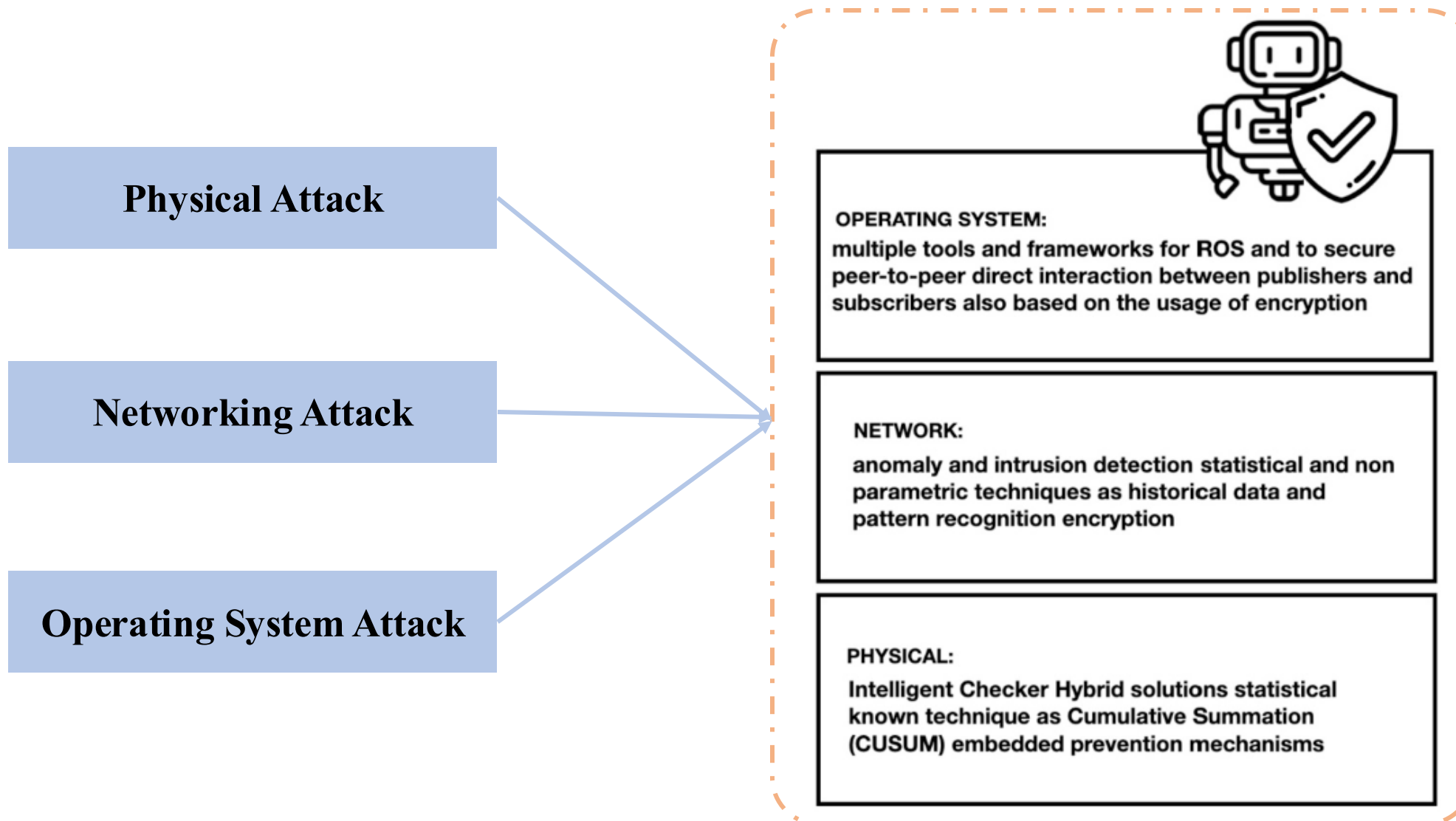
Quiz

Q3: What is a characteristic of a "Driven-by Attack"?

- A) It relies on physical access to the device for execution.*
- B) It targets network vulnerabilities in software applications.*
- C) It is a type of cyberattack where the attacker exploits vulnerabilities through external input.*
- D) It only affects hardware components of a system.*



Overall Framework





For Physical Attack

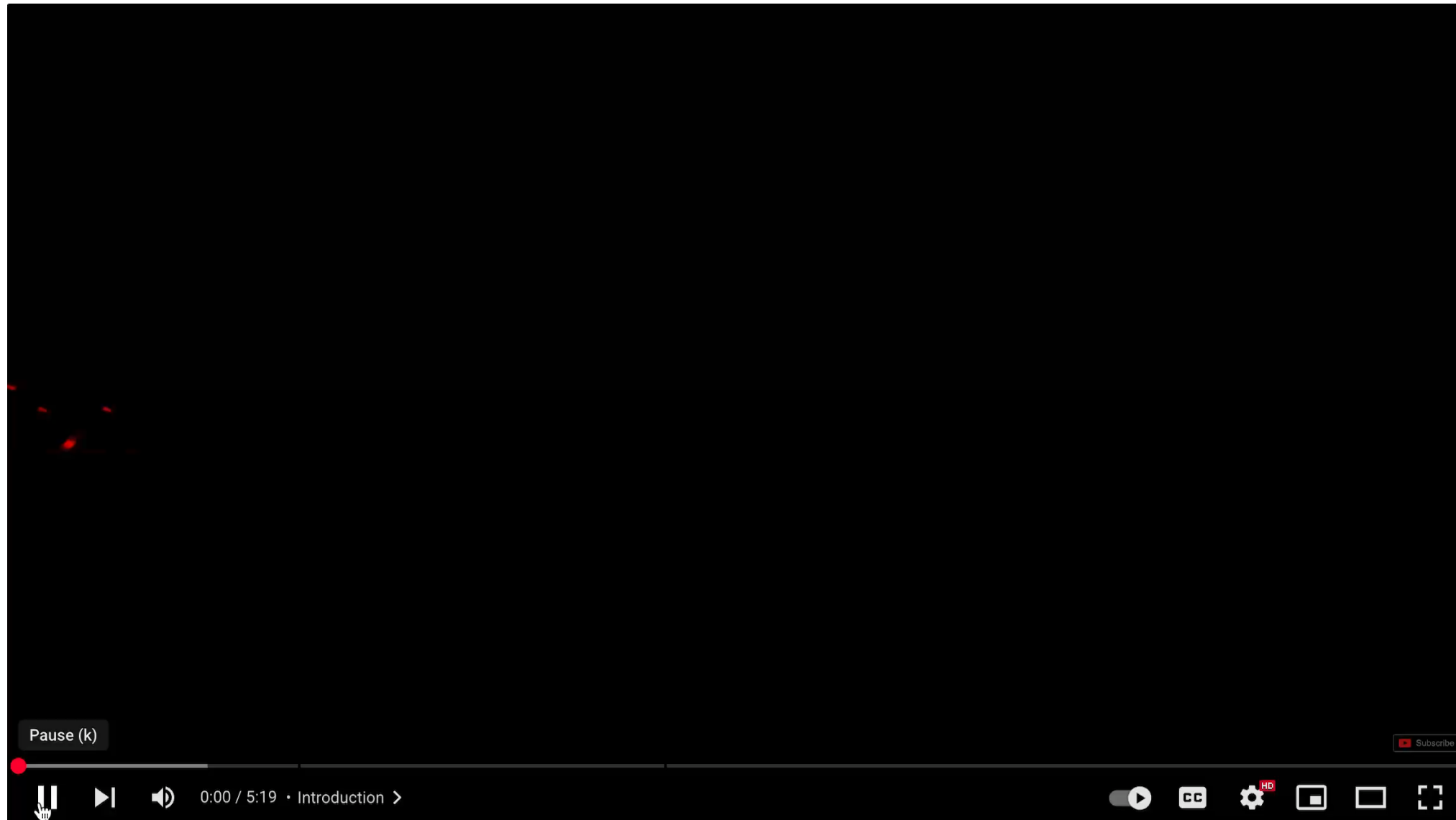
Hardware-Based Encryption: Implement hardware security modules (HSM) or secure elements directly in the robot's components. These specialized chips can protect cryptographic keys and prevent unauthorized physical access to sensitive data, even if attackers have physical access to the robot.

Redundant Sensing Systems: Use multiple sensors (e.g., LIDAR, cameras, force sensors) in different configurations to detect and respond to external physical threats. Redundant sensing ensures that if one sensor is damaged or blocked, others can still detect the attack.

Intrusion Detection Systems (IDS): Employ physical intrusion detection systems that can sense unauthorized attempts to access or damage the robot's hardware. These systems could include vibration, pressure, or motion detectors that trigger an alarm or lock the robot's components.



For Networking Attack





For Operating System Attack

- **ROSRV**, an **open-source runtime verification framework** for robotic applications aimed at addressing safety and security issues. It provides a **transparent monitoring infrastructure that monitors commands** passing through the system integrating seamlessly with RO and requiring no changes in the OS architecture
- **ROS-Defender**, a **holistic** approach integrating a **security event management system**, an intrusion prevention system and a firewall. The solution brings anomaly detection systems to the application and network level, with dynamic policy enforcement. The solution also adopts software defined networking to protect against malicious attacks



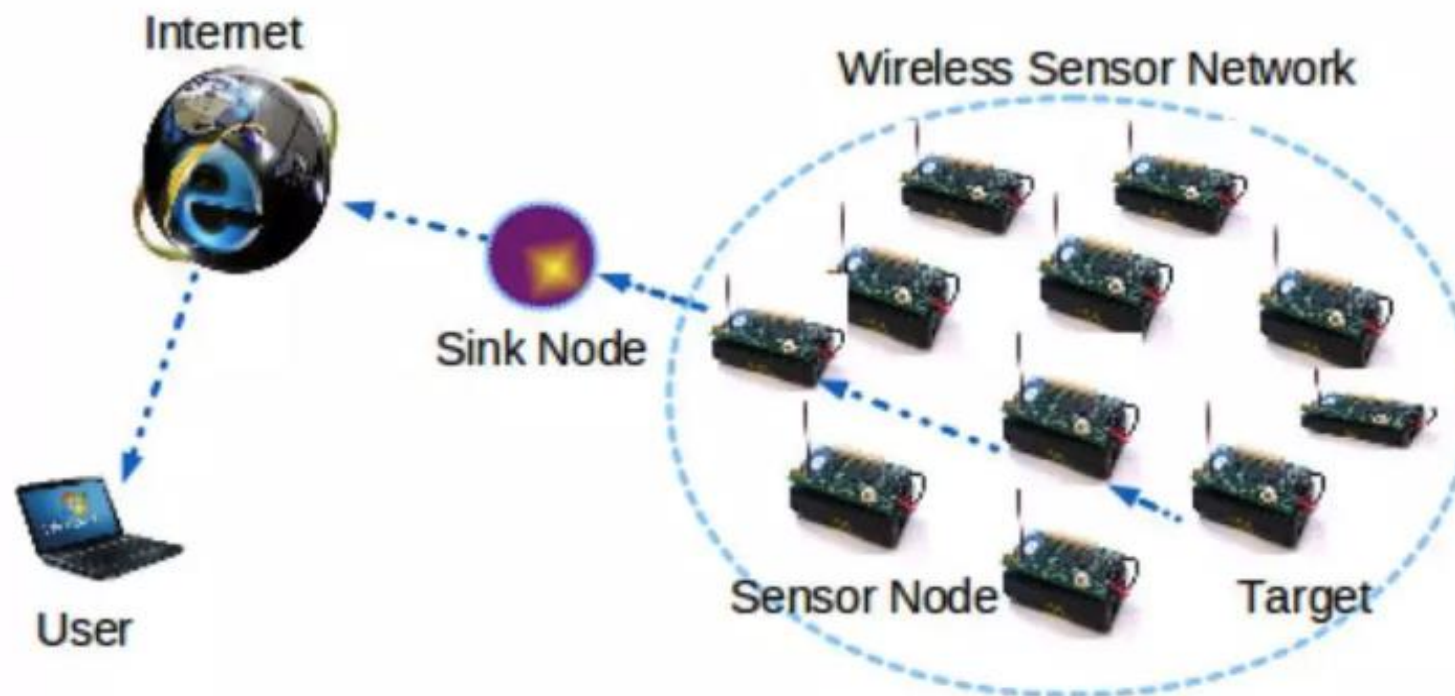
For Operating System Attack

- **ROS-Immunity**, a collection of tools, aimed at developing a modular, secure framework for ROS, which presents **internal system defense, external system verification, and automated vulnerability detection** in a unique tool that works efficiently together with Secure-ROS
- **ROS-FM**, an extended version of Berkely Packet Filters and eXpress Data Path to build a **network-monitoring framework** for ROS. When comparing the overhead of these tools to the existing one, with **more than 10 running processes** the tool appears to be causing **only 4%** of overhead with respect to other solutions



Role of Sensors in CV and Robotics

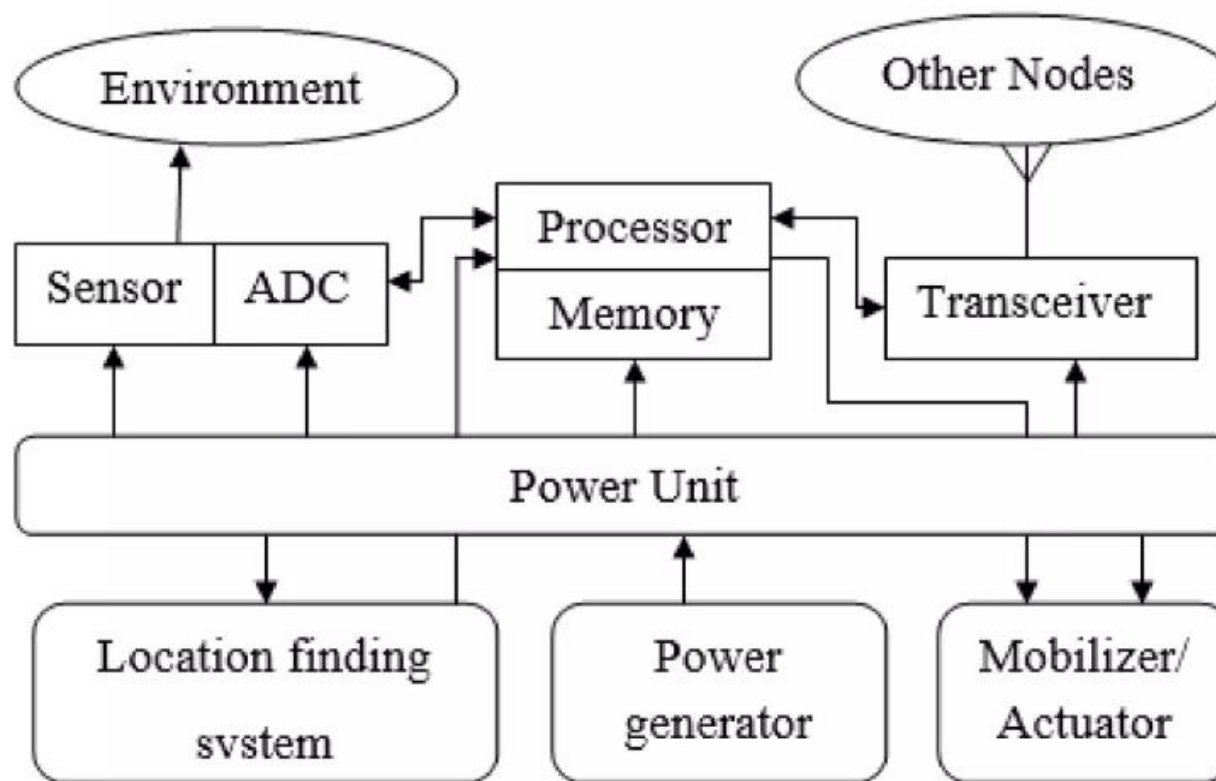
- A wireless sensor network is a wireless network consisting of **a large number of spatially distributed sensor nodes**.
- **Very sensitive** to the environment.
- **Capable of communication** with each other through wireless channel.





Role of Sensors in CV and Robotics

Architecture of Sensor Node





Role of Sensors in CV and Robotics

Introducing the sensors.

(a) Single-sensor: 360 cameras^[1] for the *comprehensive perception*



(b) Multi-sensor systems: Depth, Event, and LiDAR sensors, etc, for the *robust performance*



[1] Ai H, Cao Z, Zhu J, et al. Deep learning for omnidirectional vision: A survey and new perspectives[J]. arXiv preprint arXiv:2205.10468, 2022.



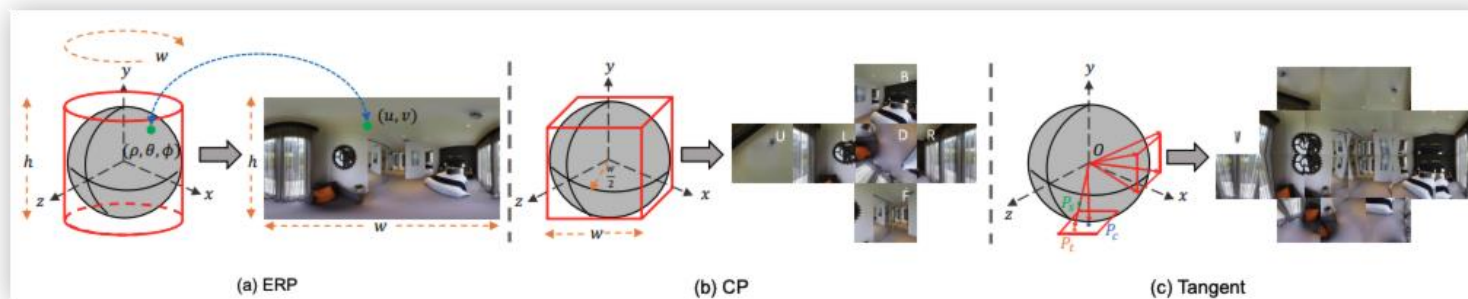
Role of Sensors in CV and Robotics

Introducing the sensors.

Novel sensors: 360 camera^[1]



(1) Equirectangular projection (ERP) of Sphere Data



(2) Different projection methods of sphere Data

[1] Ai H, Cao Z, Zhu J, et al. Deep learning for omnidirectional vision: A survey and new perspectives[J]. arXiv preprint arXiv:2205.10468, 2022.



Role of Sensors in CV and Robotics

Introducing the sensors.

Multiple sensors: Depth, Event, and LiDAR sensors^[2]



RGB

GT

Depth

Event

LiDAR



Security Risks of Sensor Data: **Sensor failures**

Sensor failures in real-world scenarios often lead to

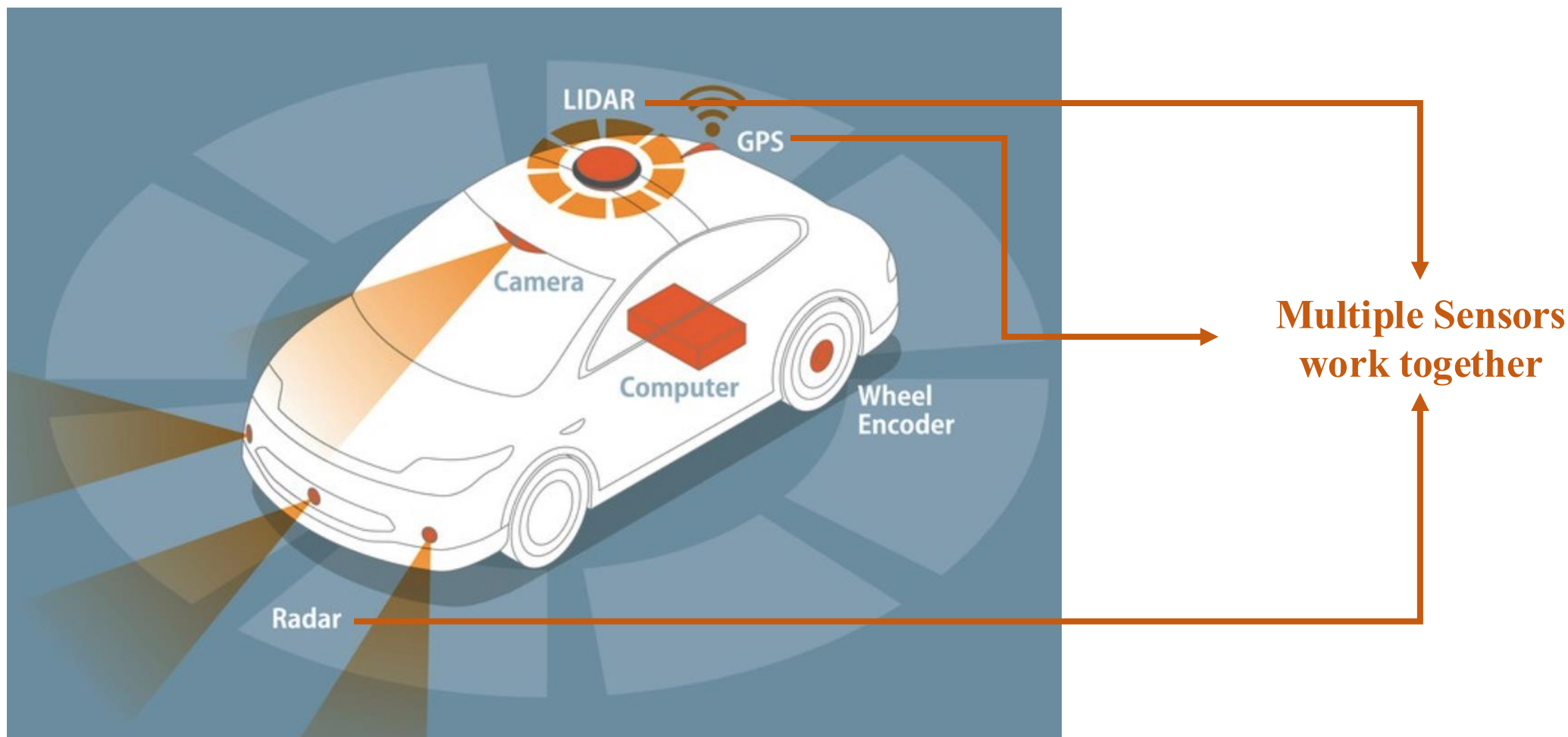
incomplete multi-source data



False Prediction



Security Risks of Sensor Data: Sensor failures



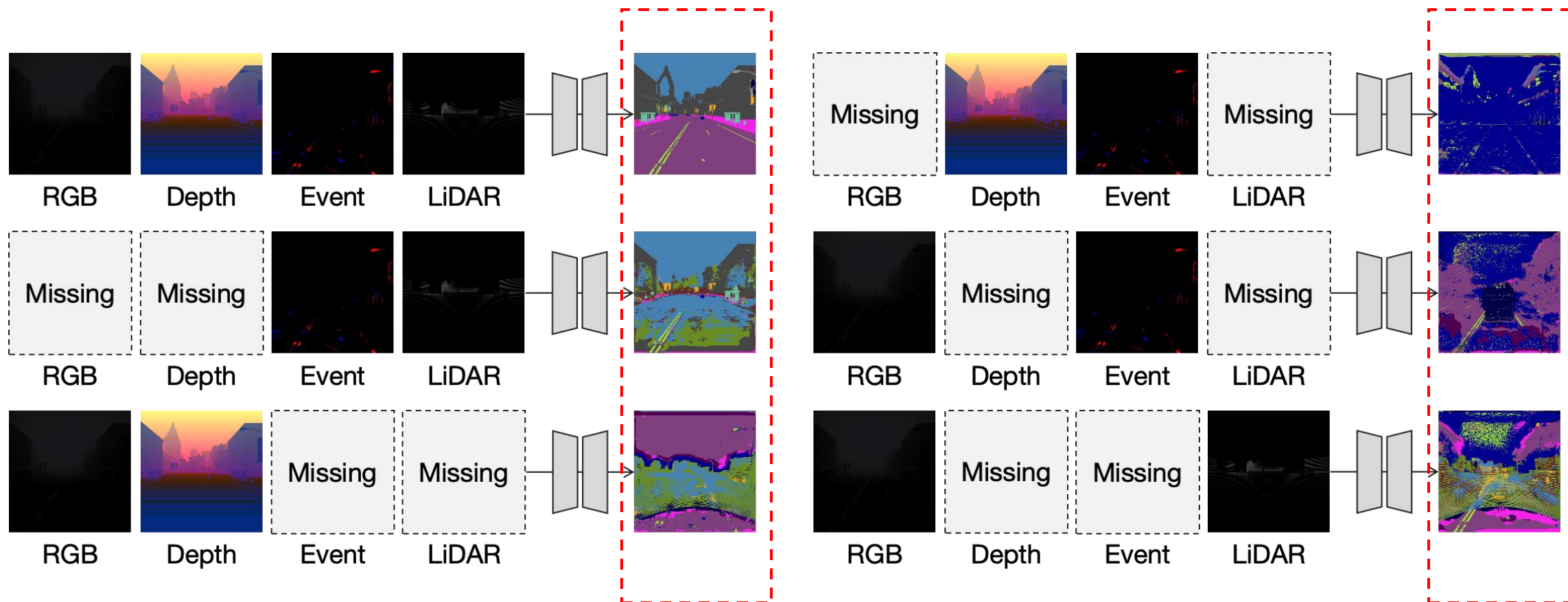


Security Risks of Sensor Data: Sensor failures

*What happens to the whole **CV&Robotics** system if one or more **sensors suddenly fail**?*



Security Risks of Sensor Data: Sensor failures



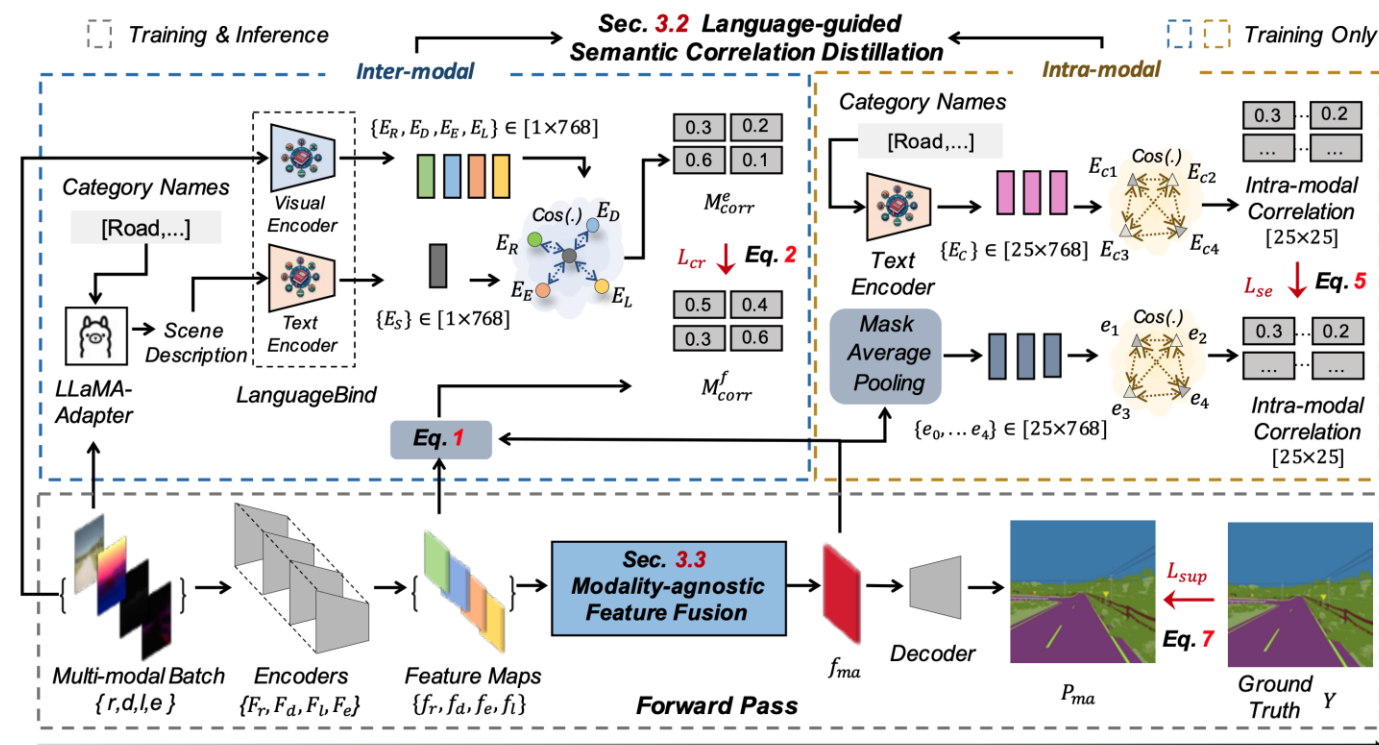


Security Risks of Sensor Data: Sensor failures

Solution:

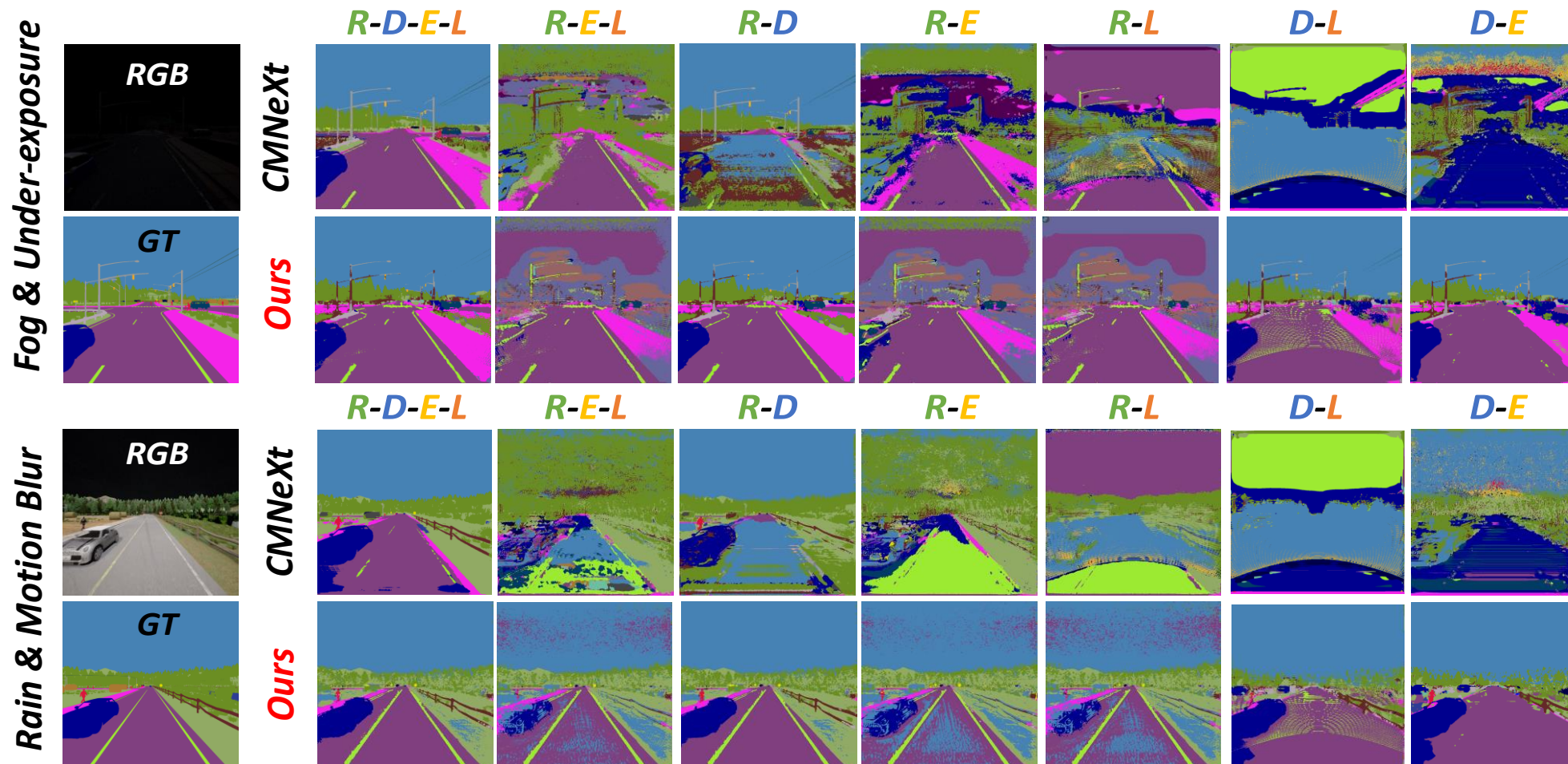
Any2Seg^[4]:

- (1) Inter-modal correlation distillation;
- (2) Intra-modal semantic distillation;
- (3) modality-agnostic feature fusion.





Security Risks of Sensor Data: Sensor failures



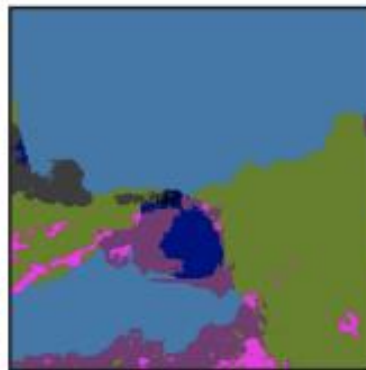


Security Risks of Sensor Data: **Sensor failures**

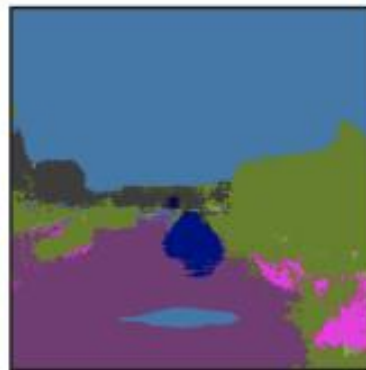
Full Sensors



Miss 4 Sensors



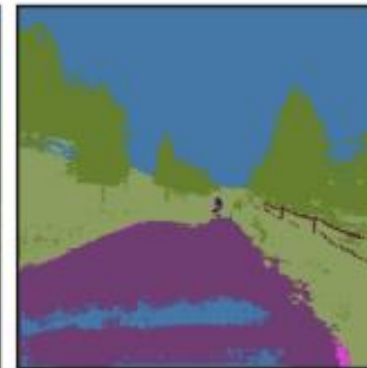
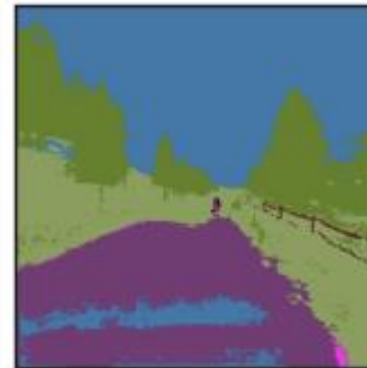
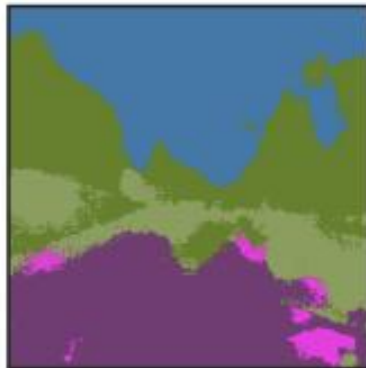
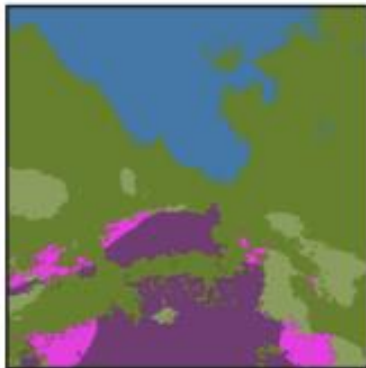
Miss 3 Sensors



Miss 2 Sensors



Miss 1 Sensors





Objective:

Use Hugging Face's *Depth Anything* demo to generate a depth map from an RGB image, and compare the differences between the two.

Steps:

1. Visit the *Depth Anything* demo on Hugging Face.
2. Upload an RGB image.
3. Generate the depth map and observe the result.
4. Compare the RGB image with the depth map. Notice how the depth map shows relative distances (closer objects lighter, farther objects darker).

Questions:

- How does the depth map represent distance differently from the RGB image?
- Why do we use both RGB and Depth sensors and how do they complement each other?



Practice: Differences in Different Sensor Data



THE HONG KONG
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AND TECHNOLOGY
(GUANGZHOU)



THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY



RGB Camera

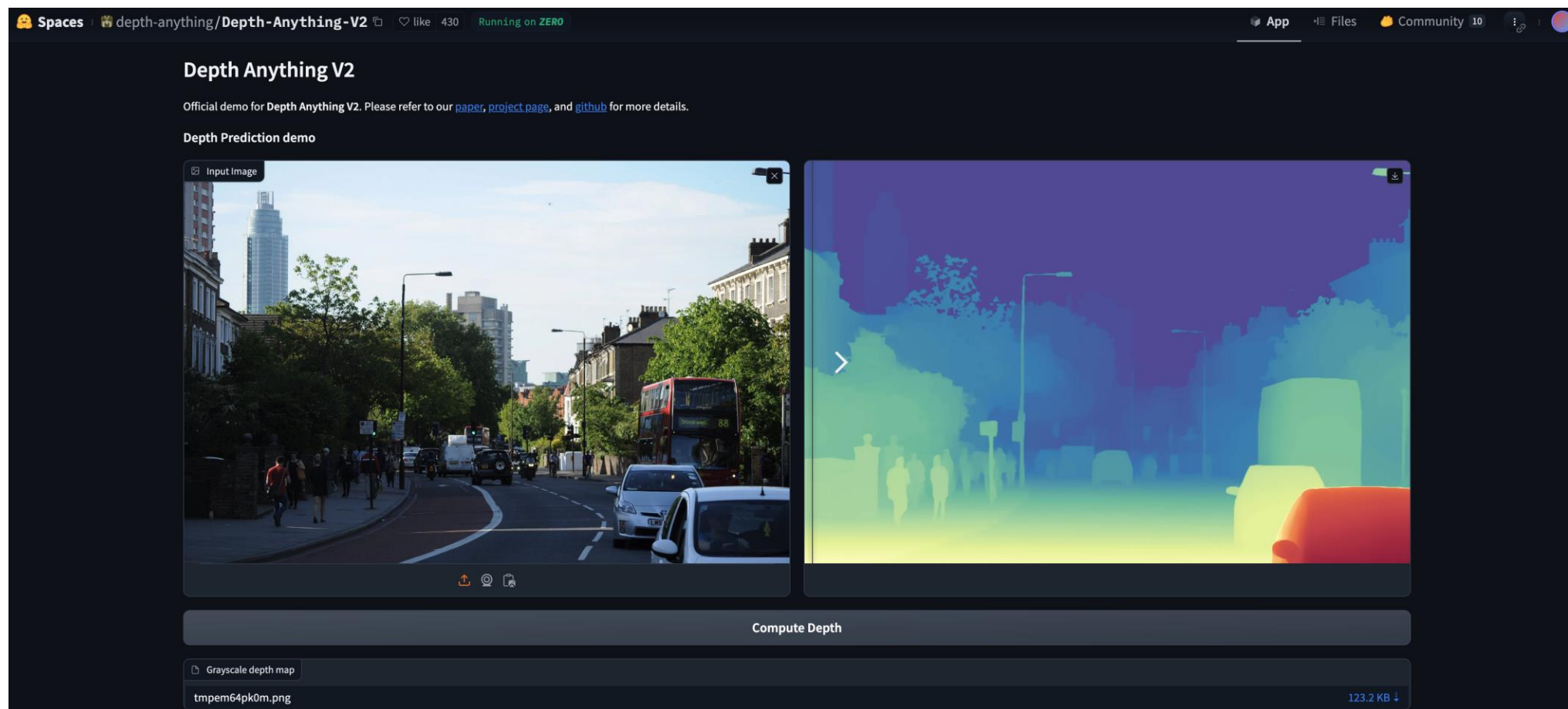


Depth Camera



Depth Anything V2:

<https://huggingface.co/spaces/depth-anything/Depth-Anything-V2>



AIAA 2290: Ethics, Privacy and Security in AI

Thanks!!

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The Hong Kong University of Science and Technology (Guangzhou)

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